Methods of proof – 6

Contradiction and proofs of impossibility

Homework

15 сентября 2026 года

Theoretical minimum. A direct proof, a proof of the contrapositive $\neg B \Rightarrow \neg A$ and a proof by contradiction are three different things; if the contradiction you reach is exactly the negation of the hypothesis, you have written a contrapositive proof with a superfluous negation. Negation with quantifiers reverses them: $\neg\forall x\,\exists y\,P$ is $\exists x\,\forall y\,\neg P$, and most failed arguments start from a wrong negation. For statements "no object with property P exists" contradiction is honest; for existence statements it is a choice which costs the example (Methods of proof – 5). Impossibility is proved by an invariant (Methods of proof – 4), by parity or a colouring, by counting or dimension, by descent (Methods of proof – 2), by the extremal choice (Methods of proof – 3), or by mapping the problem into a smaller structure — residues, determinant, degree, sign, valuation. For irrationality the standard scheme is to assume $\alpha = p/q$ and produce an integer N with $0 < |N| < 1$; a constructible number has degree a power of two over $\mathbb{Q}$, which is why the cube cannot be doubled and the angle cannot be trisected.

---

1. Show that there is no arithmetic progression with three terms inside the geometric sequence $\{2^k\}_{k\geqslant 0}$. (VJIMC)

2. Consider unit cubes in $\mathbb{R}^n$ with edges parallel to the axes and pairwise disjoint interiors. Prove that among any $2^n + 1$ such cubes some two are disjoint — or, in the equivalent form, prove that $2^n + 1$ such cubes cannot all pairwise touch; show that 2^n pairwise touching cubes exist. (МФТИ)

3. Show that there does not exist a function $f\colon \mathbb{R} \to \mathbb{R}$ with the following property: for every convergent sequence (a_i) of pairwise distinct real numbers the sequence $(f(a_i))$ converges to no real number. (IMS 2022)

4. Prove that a plane figure which has been cut into n squares with sides $1, \frac{11}{10}, \ldots, \left(\frac{11}{10}\right)^{n-1}$ parallel to the axes cannot be cut into fewer than n rectangles. (МФТИ)

5. Let a, b, c be integers such that $b^2 - 4ac$ is not a perfect square. Prove that for every $n > 1$ there are n consecutive natural numbers none of which can be written as $(ax^2 + bxy + cy^2)^z$ with integers x, y and an integer $z > 0$. (Proposed problems)

6. Prove that a parallelepiped with a base in the plane Oxy cannot be cut into tetrahedra each of which has a face parallel to Oxy. (МФТИ)

7. Do there exist two commuting linear operators on $\mathbb{R}^3$ such that no basis puts both of their matrices into Jordan form? (Мехмат 2006)

8. Prove that no continuous injection $[0,1]^2 \to \mathbb{R}$ exists, and that no continuous bijection $[0,1] \to [0,1]^2$ exists, although a continuous surjection does (Topology of the real line – 3). Identify in each case what exactly is the invariant that makes the object impossible.

9. Prove that the numbers $\sqrt[3]{2}$ and $\cos 20^\circ$ have degree 3 over $\mathbb{Q}$, and deduce that the cube cannot be doubled and the angle of 60° cannot be trisected by ruler and compass. State precisely the theorem about degrees of constructible numbers that you are using.

10. Find the error in the following argument and classify it: "Suppose $\sum 1/n$ converges to S. Grouping the terms in pairs gives $\sum(1/(2k-1) + 1/(2k)) > \sum 1/k = S$, hence $S > S$, a contradiction; therefore the harmonic series diverges." Then give a correct proof by contradiction of the same statement, and a correct direct proof, and say which one you would submit.